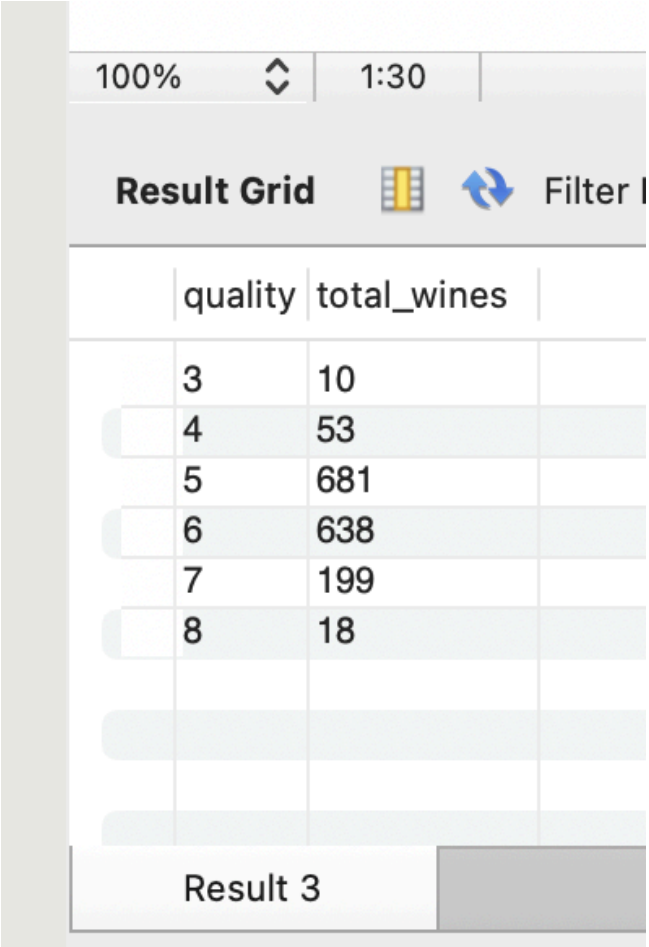
```
1.SELECT
quality,
COUNT(*) AS total_wines
FROM wine_quality_red
GROUP BY quality
ORDER BY quality;
```



quality	total_wines
3	10
4	53
5	681
6	638
7	199
8	18

Figure 1

Most wines are concentrated in the mid-quality range (scores 5 and 6), indicating that the dataset is heavily centered around average-quality wines, with very few low- or high-quality outliers.

2. SELECT
quality,
ROUND(AVG(alcohol), 2) AS avg_alcohol
FROM wine_quality_red
GROUP BY quality
ORDER BY quality;

100% 18:37

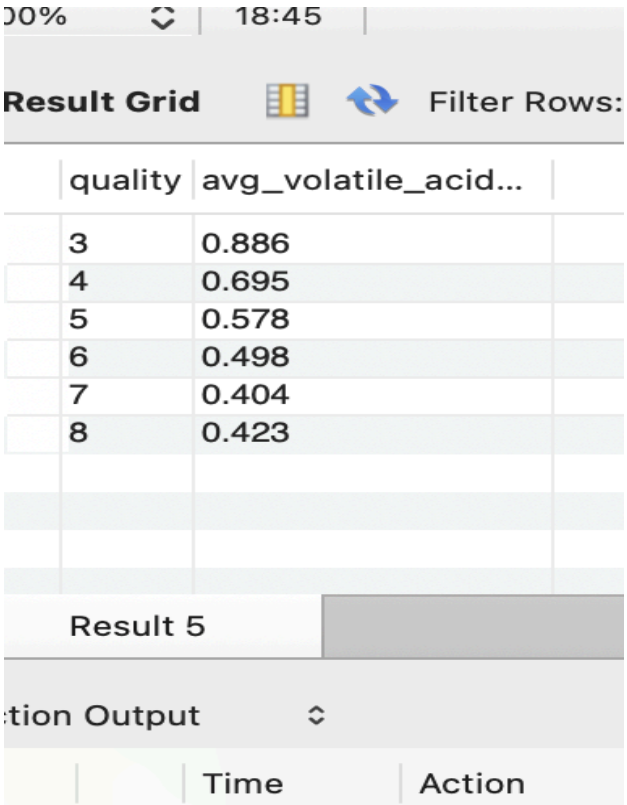
Result Grid Filte

quality	avg_alcohol
3	9.96
4	10.27
5	9.90
6	10.63
7	11.47
8	12.09

Figure 2

There is a clear positive relationship between alcohol content and wine quality. Higher-quality wines tend to have significantly higher average alcohol levels, suggesting alcohol content may be a strong indicator of wine quality.

3. SELECT
quality,
ROUND(AVG(volatile_acidity), 3) AS avg_volatile_acidity
FROM wine_quality_red
GROUP BY quality
ORDER BY quality;



quality	avg_volatile_acid...
3	0.886
4	0.695
5	0.578
6	0.498
7	0.404
8	0.423

Figure 3

Volatile acidity shows a strong negative relationship with wine quality. As quality increases, volatile acidity consistently decreases, indicating that lower acidity levels are associated with better-quality wines.

4.SELECT
quality,
ROUND(AVG(sulphates), 2) AS avg_sulphates
FROM wine_quality_red
GROUP BY quality
ORDER BY quality;

quality	avg_sulphates
3	0.57
4	0.60
5	0.62
6	0.68
7	0.74
8	0.77

Result 6

Action Output

Figure 4

Sulphate levels increase with wine quality, suggesting that higher sulphate concentration may contribute to improved wine stability and overall quality.

5. SELECT *
FROM wine_quality_red
ORDER BY quality DESC, alcohol DESC
LIMIT 10;

100% 56:55

Result Grid Filter Rows: Search Export: Fetch rows:

fixed_acidi...	volatile_acid...	citric_ac...	residual_sugar	chlorides	free_sulfur_dioxi...	total_sulfur_dioxi...	density	pH	sulphates	alcohol	quality
5.50	0.49	0.03	1.80	0.044	28.00	87.00	0.9908	3.50	0.82	14.00	8
11.30	0.62	0.67	5.20	0.086	6.00	19.00	0.9988	3.22	0.69	13.40	8
7.90	0.54	0.34	2.50	0.076	8.00	17.00	0.9924	3.20	0.72	13.10	8
5.60	0.85	0.05	1.40	0.045	12.00	88.00	0.9924	3.56	0.82	12.90	8
7.90	0.35	0.46	3.60	0.078	15.00	37.00	0.9973	3.35	0.86	12.80	8
7.80	0.57	0.09	2.30	0.065	34.00	45.00	0.9942	3.46	0.74	12.70	8
10.30	0.32	0.45	6.40	0.073	5.00	13.00	0.9976	3.23	0.82	12.60	8
9.10	0.40	0.50	1.80	0.071	7.00	16.00	0.9946	3.21	0.69	12.50	8
10.00	0.26	0.54	1.90	0.083	42.00	74.00	0.9945	2.98	0.63	11.80	8

wine_quality_red 7

Action Output

Figure 5

The highest-quality wines in the dataset (quality score of 8) consistently exhibit higher alcohol content and relatively lower volatile acidity levels. Additionally, these wines tend to have moderate to higher sulphate concentrations. This suggests that premium wines are typically associated with stronger alcohol presence, better chemical balance, and lower acidity.

6Question: How do key chemical properties differ between low, medium, and high-quality wines?

```
SELECT
   CASE
      WHEN quality >= 7 THEN 'High Quality'
      WHEN quality = 6 THEN 'Medium Quality'
      ELSE 'Low Quality'
   END AS quality_category,
   COUNT(*) AS total_wines,
   ROUND(AVG(alcohol), 2) AS avg_alcohol,
   ROUND(AVG(volatile_acidity), 3) AS avg_volatile_acidity,
   ROUND(AVG(sulphates), 2) AS avg_sulphates
FROM wine_quality_red
GROUP BY quality_category;
```

100% 1:74

Result Grid Filter Rows: Search Export:

quality_catego...	total_wines	avg_alcohol	avg_volatile_acid...	avg_sulphates
Low Quality	744	9.93	0.590	0.62
Medium Quality	638	10.63	0.498	0.68
High Quality	217	11.52	0.406	0.74

Figure 6

High-quality wines are characterized by higher alcohol content, lower volatile acidity, and higher sulphate levels. In contrast, low-quality wines tend to have lower alcohol content and higher acidity, indicating that these chemical properties play a significant role in determining wine quality.